

The PAMalyzer (v0.0.1)

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This manual must be considered work in progress and still needs to be updated!

This is the user manual for ▲ version 0.0.1 of the PAMalyzer program. The PAMalyzer is based on AviaNZ, a software for bioacoustic analysis (<http://www.avianz.net>). There are so many changes compared to AviaNZ that I decided to name it differently. The most obvious are (i) the ability to classify your recordings with the BirdNET classifiers, and (ii) several features to ease the analysis of the BirdNET results with filtering and sorting options. The purpose of this manual is to help you get started with using the PAMalyzer. Furthermore a 'Cheat Sheet' is provided which you can find in the same directory as this manual or within the program under Help.

Quick-Start

1. **Set operator:** On first startup PAMalyzer will ask you to enter an operator name for future reference. More information in [Getting Started](#).
2. **Open an audio file:** If you did not use PAMalyzer before an example file will be loaded. You can use the *Open sound file* option in the *File* menu or press **Ctrl+O**. See [Open Audio Files](#) for details.
3. **Choose a database:** All annotations in PAMalyzer are stored in a database. The default database will be good for first tests. Choose another database via *Annotations>Database>Choose database* if you start productive work (see the [Database selection](#) section for more information).
4. **Classify recordings:** To classify your audio files launch BirdNET via *Analysis→Classify recordings with BirdNET*. Find more detailed information in the [Analyze Recordings with BirdNET](#) section.
5. **Review detections:** PAMalyzer offers plenty of options to filter and review detections. How to annotate recordings is described in [Annotate Recordings](#). Filtering options are listed in the section [Sort and Filter the List of Files](#).

6. **Export results:** Export annotations and summary Excel files via the *Export* options in the *Annotations* menu. Refer to the [Export](#) section for the export formats.

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1 Introduction

PAMalyzer is free and open-source software that features:

- Display of spectrograms and quick navigation through audio files
- Manual labeling of sounds in audio files
- Classification using different BirdNET-based models (BirdNET Lite, BirdNET v2.4, custom models based on BirdNET)
- Quick manual review of classification results using audio replay and spectrogram display

If you are using PAMalyzer the first time the section [Getting Started](#) is a good starting point.

2 Installation and Setup

For Windows users there is a downloadable installation file available on the GitHub page <https://github.com/FloMee/PAMalyzer/releases>. This should work for most users.

If you plan on using the software under Linux or MacOS, follow the instructions in the README to install the software from source.

***Note:** On Windows some files will be placed in your home directory, which may be somewhere like `C:/Users/username/AppData/Roaming/PAMalyzer`. On Mac and Linux they will be in `~/PAMalyzer`.*

3 Getting Started

On the first startup the program will ask you to give a name for the operator. This is useful to keep track of who has looked at different files. After confirming the main interface (figure 1) shows up.

The interface of PAMalyzer is structured in 4 separate components. Each has its name in a blue bar. They are:

Overview (1) This area shows a spectrogram of the first 5 minutes of the current audio file and gives an overview of annotations in the file. The blue section indicates the part you are looking at in the spectrogram plot below. You can use the overview spectrogram to quickly identify sections of interest and navigate to them. The options you have for navigating through audio files are listed in section [Navigate through Files](#). All the components of the overview area are described in detail in section [9.1.1](#).

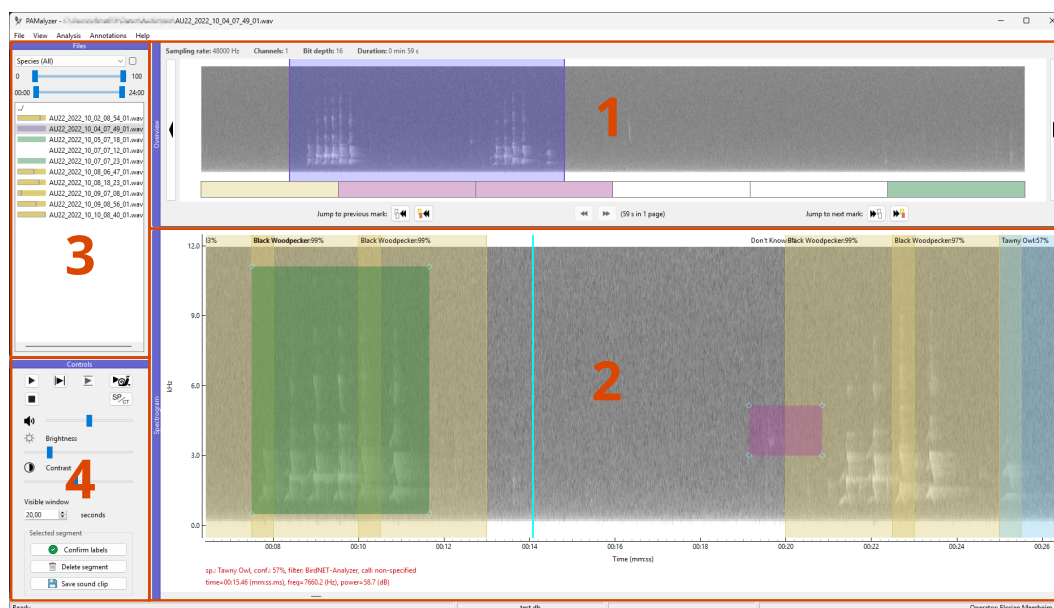


Figure 1: The interface of PAMalyzer with its different components.

Spectrogram (2) In this area the spectrogram of the blue section in the Overview area is shown. Here you can add and modify annotations. More information on the spectrogram settings and the way in which you annotate recordings is described in the sections [Spectrogram](#) and [Annotate Recordings](#) respectively. A detailed description of all the components in the spectrogram area is given in section [9.1.2](#).

Files (3) This is the list of audio files in your current directory along with filtering options. Use the **Up** or **Down** arrow buttons on your keyboard to navigate through the file list. You can also double-click on a file to select and open it (more details in section [Navigate through Files](#)). The file icons next to the file names indicate the reviewing status of the audio file. This color coding is described in detail in section [5.5](#). The filtering options above the file list are described in section [Filtering](#). A reference for all the components can be found in section [9.1.3](#)

Controls (4) The buttons in this area are referring to the Spectrogram area. Use the buttons here to playback the audio file, modify the appearance of the plots and process the currently selected segment. For more details, see [Controls](#).

There is an optional fifth area, which is the Amplitude plot. If you want to see it, select *Show amplitude plot* in the *View* menu. Select the option again to hide it. You can also hide the *Files* and *Overview* area in the same way. PAMalyzer will remember your choice in future sessions.

There are two other parts to the interface:

Menu The menu at the top of the screen which gives access to most of the options in PAMalyzer. A detailed description of all the menus is given in section [9.2](#).

***Note:** You can drag the five screen areas around and reorder them if you wish, by dragging the blue bar on the top or left of them with the name of the section in it. You can also make them into their own windows by double-clicking on the blue bar. If you decide that you made a mistake doing that, then there is an option in the View menu to Restore default layout that returns them to the original configuration.*

Status Bar At the very bottom of the screen there is an area that shows the current operator on the right along with the currently selected database in the middle and some other status information.

3.1 Cheat Sheet

You can find a Cheat Sheet with the most important navigation options, especially shortcuts in the *Help* menu under *Cheat Sheet*.

4 Open Audio Files

By default a demo audio file is shown. If you want to move to a new directory, either use the ‘..’ option to navigate around your computer’s file system, or use the *Open sound file* in the *File* menu.

***Note:** PAMalyzer can load audio files in wav format. Other audio files should be converted to wav format. PAMalyzer assumes that audio files are recordings in mono (single microphone) sound. If there are other channels, only the first one is loaded. Very long recordings can be used, but it is generally easier if they are split into shorter files.*

4.1 Spectrogram

4.2 Navigate through Files

To quickly navigate through the list of files and the files themselves it is highly recommended to use the arrow keys on the keyboard: Use the **Up** and **Down** arrow keys to open the previous or next audio file in the file list. Use the **Left** and **Right** arrow keys to move backward and forward through the audio file.

But there are many more options

- To move between audio files:
 - Use the **Up** or **Down** arrow keys to open the next or the previous file in the file list.
 - Use *Open audio file* in the *File* menu if you want to open an audio file in another directory.

- Double Click on a file in the list of files.
- Double Click on a folder will open that folder; the ‘..’ option at the top of the list takes you up one directory.

Note: *If some of the filter options in the Files area (10) are applied, some of the files might be hidden within the file list. In this case the Up and Down arrow keys apply only for the visible files. The shortcut Alt + Up/Down in contrast includes also the hidden files, so that you can still reach them.*

- To move within a file:
 - Use the **Left** or **Right** arrow keys to move forward or backward within the file.
 - Drag the blue highlighted area in the overview spectrogram.
 - Click on the left or right arrows on the left and right of the overview spectrogram (3).
 - Click on any of the boxes in the bar below the overview spectrogram (4).
 - Drag the scroll bar below the spectrogram.

Note: *According to the set page size (defaults to 5 minutes) in the [Interface settings](#) long files are split into several pages. To open the next or previous page of an audio file use the double arrow buttons (5) below the overview spectrogram or the shortcut **Shift + Left/Right**. More details in: [Moving Through Long Files](#).*

- To change the amount of the file that you can see (the visible width):
 - Drag the ends of the blue highlighted section in the overview spectrogram.
 - Change the *Visible window width (seconds)* in the Controls dock by clicking on the up and down arrows, or typing in a new number.
- You can also view a restricted amount of the spectrogram by reducing the visible frequency band, see Section [Changing the Spectrogram Computation](#).

Moving Through Long Files According to the set page size (defaults to 5 minutes) in the [Interface settings](#) long files are split into several pages.

- To move to the next or previous page:
 - Use the double arrow buttons labeled (5) or
 - press **Shift + Left/Right** arrow keys.
- The program tells you which page you are currently on, and how many there are in total.

- You can change the page size and the amount of page overlap by choosing the relevant options in the *Interface settings* in the *View* menu.
- The times on the axis below the spectrogram show locations in the full file.
- Note that operations like denoising and segmentation (described in the section *Analysis Menu*) apply to the visible 5 minute portion of the file, not the whole file.

5 Annotate Recordings

5.1 Overview

The aim of annotating audio files is to add further information. For this purpose PAMa-lyzer uses labeled segments that are stored in a database. A segment is a specific range in time and frequency within the spectrogram. Together with the information about time and frequency a species label (e.g. a bird species) and a call type label (for specific call types of the species) can be stored.

- To create segments, click and drag with the left mouse button on the spectrogram. (The action of left and right mouse buttons can be swapped over in the *Interface settings*.)
- To select segments, click on them with the right mouse button (pressing control on the keyboard when clicking on a Mac). The segment will turn blue when it is selected.
- If you find that the colors make it hard to see the data underneath the boxes you can make them transparent using the *Make boxes transparent* option under *Annotation* in the *Interface settings*. You can also choose different colors there if you have particular preferences.
- Segments are saved automatically in the *Database selection*.

5.2 Database selection

Some important notes:

- Choose a database via *Database -> Choose database*. You can either choose an existing one or create a new one.
- Within the database file names are used as primary key but
- Queries are based on the directory in most cases.
- Update the file directory if you move a file to another directory via *Database -> Update file directory*.

Note: *If many thousands of annotations are stored in the database PAMalyzer starts to perform worse. To keep the performance of PAMalyzer high, project related databases are recommended. You can also export subsets of annotations to new databases using the Export functions in the **Annotations** menu.*

5.3 Create and Label Segments

- There are three ways to create a segment. You change which of them to use in the *Mouse settings* of the *Interface settings* in the *View* menu:
 - (default) Drag a limited frequency band box (i.e., click and hold the mouse button and drag the mouse to the correct end point in both time and frequency).
 - Start and stop a limited frequency band by clicking (i.e., click once at the correct time and frequency for the start of a segment, and then again at the time and frequency for the end).
 - Start and stop a full frequency band by clicking (i.e., click once at the correct time for the start of a segment, move the mouse to the end, click again, the box covers all frequencies).
- When you create a new segment, a drop-down menu will appear asking you to choose a label for that segment.
 - To choose the type of bird, click on the name.
 - If the species isn't in the list, move to 'Other'.
 - If it isn't in the second list, at the bottom of that list is a selectable box (it probably says 'Albatross'). Clicking on it provides the complete list of birds.
 - If there is something missing from there, choose 'Other' from it, which is at the very end.
 - The new name you add will then appear in the list.
 - If you click anywhere on the screen except on a bird name in the menu then the menu will disappear and the bird type will be labeled as 'Don't Know'.

Note: *The original version of PAMalyzer comes with a bird list for New Zealand. In the list the species names are given as 'Genus (Species)'; e.g. 'Kiwi (Little Spotted)'. Given it in that way a submenu for every genus is created with the species in it. Although this works great for common species names in English, it doesn't in other languages. It is possible to use other bird lists (that do not necessarily have to follow this formatting) using the *Interface settings*.*

- By default the lists of bird names updates dynamically, so that bird types you have chosen appear at the top of the list. If you don't like that, then you can disable it in the *Interface settings*.

- When creating a segment, you can give it the same label as the previous segment by pressing the **Shift** key on the keyboard when you create the segment. The program will then give the segment the same label as the previous box that you labeled, without showing you the list. This is very useful when there is one bird calling repeatedly.
- You can also show that you are uncertain by pressing the **Ctrl** button when you click to segment (command button on a Mac computer). The names of the birds will then have a question mark after them. Internally this adds a confidence score of 50 to the annotation. Do not use this option if you are working with BirdNET.
- You can choose whether or not to allow more than one species per segment. This can be useful for recordings of the dawn chorus and where you just want to label presence of a set of species. Use the *Default to multiple species* in the [Interface settings](#) to allow this. Then, when you choose a bird from the list it will be ticked, and the menu will not close automatically, allowing you to make multiple selections. To deselect something, click on it again. When you make a new segment, ‘Don’t Know’ is selected by default. Choosing any other option deselects ‘Don’t Know’. You can reselect it if you do want that label as well.
- PAMalyzer also allows you to specify a call type. This could be the difference between male and female vocalizations for e.g., kiwi, or different calls types such as ‘more-pork’ or ‘cree’ for ruru. You can add specific call types manually. To swap between seeing the species and seeing the call type label use the SP/CT button () on the right of area 12 of the Controls panel or press the **Tab** key.

5.4 Confirm/Change/Correct segments

To manipulate existing segments, different options exist:

- To confirm the label of a selected segment use the *Confirm labels* button in the *Controls* area or simply hit the **Return** key.
- Use your mouse to drag single segments to change their position.
- Use your mouse and the diamonds in the corners of a selected segment to resize it.
- Use the *Delete segment* button or the key **Backspace** or **Del** to delete segments
- Use the *Delete all segment* option from the *View* menu or the shortcut **Ctrl + D** to delete every segment in the current file. This can be handy if you want to replace a lot of BirdNET detections in a file with just one segment.
- If you can neither confirm nor refuse the label BirdNET has given to the segment it might be helpful to assign a custom ‘species’ label. Later, this label can be used to filter out recordings with these doubtful segments. The process how you can add custom labels is described in [Create and Label Segments](#).

Note: *If there are several species attached to a single segment you can only confirm all the labels. In that case confirm all the labels first and deselect the “wrong” species by simply clicking with the right mouse button on the segment and deselecting the species from the opening menu.*

- Additionally to the species label PAMalyzer provides the options to assign call type labels. You can change the type of the labels that are shown in the segments by pressing the *SP/CT* button in the controls area or with the **Tab** key. With right clicking on a selected segment a menu opens where you can select call types from a list or add a new call type for the specific species by choosing the *Add calltype* option.
- You can disable the making and updating of segments (to avoid making segments by mistake) by selecting *Make read only* from the *View* menu.

5.5 Color Codes

- The segments that are drawn on the screen have different colors. These colors are changeable in the *Interface settings*, but by default are:
 - Blue** This segment is currently selected. You can play it by pressing the buttons in the Controls area (see Section 9.1.4), or give it a new label by clicking on it again, or delete it.
 - Green** This segment has been labeled with a bird name and a confidence value of 100. These are either manually created segments or the manually confirmed results of an automatic filter.
 - Yellow** This segment has been labeled with a bird name and a confidence value between 0 and 100. These are either manually created segments with a question mark (by pressing **Ctrl** while dragging the box), or the row result of an automatic filter.
 - Red** This segment has been labeled as ‘Don’t Know’. Its confidence value is 0.
- These colors match the rectangles underneath the Overview panel (labeled 4 in figure 3). For each 10 second segment of the file, these boxes are:
 - White** if there are no segments, or
 - Red** if there are ‘Don’t know’ segments, or
 - Yellow** if there are questionable segments, or
 - Green** if all the segments in that section are labeled with definite species.
- The described color scheme also matches the colors of the file icon in the file list (labeled 16 in 5).
 - White** if there are no segments in the file or directory, or

Red if there are 'Don't know' segments, or

Yellow if there are questionable segments, or

Green if all the segments in the file or directory are labeled with definite species.

- Moreover, the gray box in the file icon indicates the maximum confidence value found in the segments of the file.

***Note:** The filtering options for the file list (10) also effect the file icon. If the segments in the file do not match the set conditions the file icons turn white. You can use this feature to quickly search for candidate recording after the classification process with BirdNET. The workflow with BirdNET is described in the section [Analyze Recordings with BirdNET](#)*

6 Analyze Recordings with BirdNET

To start the analysis of your audio files with BirdNET, select *Classify recordings with BirdNET* from the *Analysis* menu. That will open the parameter input dialog (figure 2).

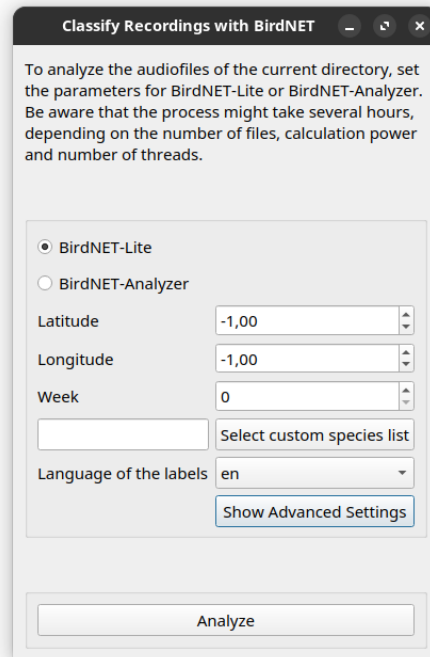


Figure 2: The BirdNET dialog

In this window the parameters for BirdNET-Lite or BirdNET-Analyzer can be set. The button *Show advanced settings* gives even more options. The meaning of the different parameters is explained in the cheat sheet that you can find over *Help → Cheat Sheet*.

Note: *The predictions of BirdNET will be stored as annotations in the currently selected database. The chosen parameters affect the number of predictions. Low values for the Minimum confidence or high values for the overlap parameter result in higher numbers of predictions. Also a custom species list can help to minimize the number of predictions.*

After the parameters are chosen, the analysis can be started by clicking on the *Analyze* button. That will classify all the recordings in the current directory including subdirectories.

6.1 The output of BirdNET

Internally BirdNET cuts your recordings into chunks of 3 seconds that overlap more or less depending on the set overlap parameter. For each of these chunks BirdNET produces a list of all classes BirdNET can distinguish (mainly bird species). To each of these species a confidence score ranging from 0 to 1 is attached. With setting the different parameters of BirdNET (for example: minimum confidence, latitude, longitude, week, custom species list) this list is reduced to a minimum of at most a few species.

For interpreting the results it is important to understand that the confidence scores are not calibrated. That leads to two important things:

- The confidence score is not a true probability or certainty value. A value of 0.8 does not mean, that on average the species can be found in 8 out of 10 segments labeled with this certainty value.
- The confidence scores can not be compared across different species. A value of 0.8 can have a completely different meaning for a Pygmy Owl than for a Capercaillie.

7 Sort and Filter the List of Files

PAMalyzer offers different options to sort and filter the file list for files and annotations. These should help to effectively review predictions from classifiers such as BirdNET.

7.1 Sorting

- By default the list of files is sorted by their file names.
- Change the sorting behavior by selecting *Sort files by maximum confidence* in the *View* menu. This sorts the list of files descending by the highest confidence scores detected in the recording for the selected species.

***Note:** Usually the recordings include a time stamp in the format Year-Month-Date-Hour-Minute-Second - or similar. Sorting the files by file name means sorting by time in these cases as well. In this mode moving to previous/following recordings in the file list also means moving to temporal adjacent recordings. This is not possible if the files are sorted by confidence.*

7.2 Filtering

Use the options above the list of files to filter recordings:

- Choose a species from the species list.
- Tick the box next to the species list to filter out recordings without annotations of the selected species
- Adjust the confidence slider below the species list to filter for recordings with annotations within the confidence range.
- Adjust the time slider to filter for recordings within the set time interval.

Use these options to systematically search for audio files with high confidence scores for the specified species (considered “candidate recordings”). A full reference with more details is given in section [9.1.3](#).

8 Export

The PAMalyzer program aims to provide detailed and easy-to-review outputs. An Excel file (with the name ‘DetectionSummary’ followed by the species selected) will be generated with two sheets of outputs in the same directory as the files. If this file already exists, the new results will be appended to the end of each sheet.

The three sheets of the Excel workbook are:

1. start and end times of each birdcall detected
2. presence/absence of the target species (or set of species) in each recording

9 Reference

9.1 Interface panels

9.1.1 Overview

The overview panel shows the first five minutes of the currently selected audio file as spectrogram along with some information on the audio file specifications and offers options to move through the file with buttons.

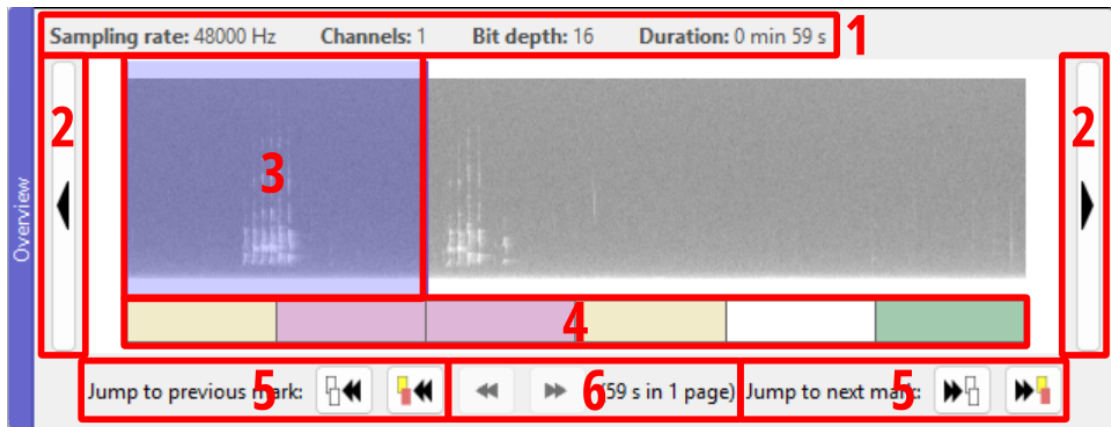


Figure 3: The Overview panel of the main interface.

- (1) Metadata of the audio file: Sampling rate in Hertz, number of channels, bit depth and duration.
- (2) Move through the file with these buttons.
- (3) The blue area is shown in more detail in the spectrogram plot below.
- (4) Jump to other parts of the file by clicking on the buttons. They also indicate if annotations are found in that section of the file. More information on color coding in section 5.5.
- (5) Jump with the blue section to the next or previous annotation segment.
- (6) Open the next or previous 5 minutes page of the audio file, provided it exists.

9.1.2 Spectrogram

The main spectrogram plot shows a section of the current audio file. The part you are looking at is highlighted in blue in the overview spectrogram (labeled 3 in figure 3).

- (7) Labels of the segments are shown, along with the confidence scores, on top of the spectrogram.
- (8) Position of the playhead: If you start playing audio it starts from here. You can 'place' it in the spectrogram by clicking. When the sound is paused, you can drag it to hear a particular part of the file. Move the mouse over it, and the bar will go red. Then click and drag (using the mouse button that does not make segments, by default the right button) to move it.
- (9) The vertical axis of the spectrogram shows frequencies.

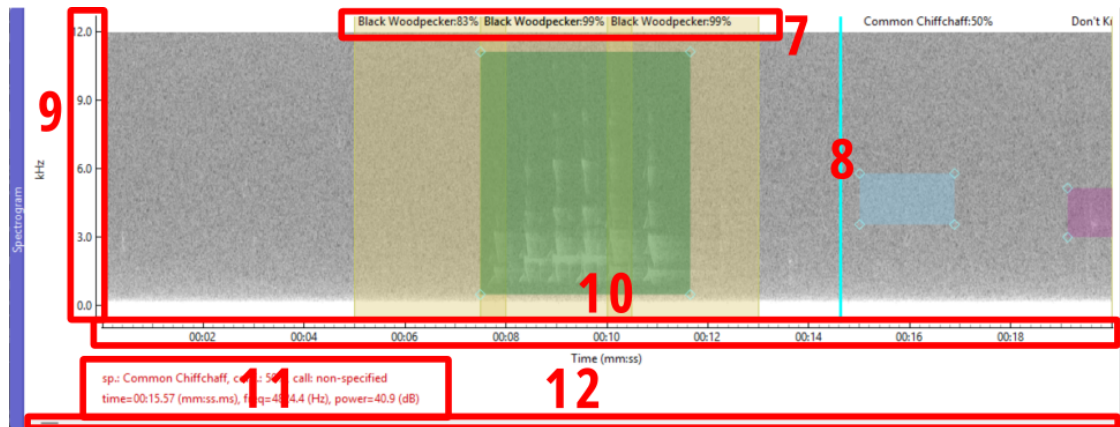


Figure 4: The Spectrogram panel of the main interface.

- (10) The horizontal axis shows time. If the file name contains date and time in the format *<year><month><day>_<hour><minute><second>* they will be true times, or time from start of file otherwise.
- (11) Detailed Information on the currently selected segment and the mouse position in the spectrogram (if selected from *View – Show pointer details in spectrogram*) is shown here.
- (12) A horizontal scrollbar that offers the option to move through the file.

9.1.3 Files

The files panel lists all the audio files and folders in the currently selected directory and offers options to filter the file list.

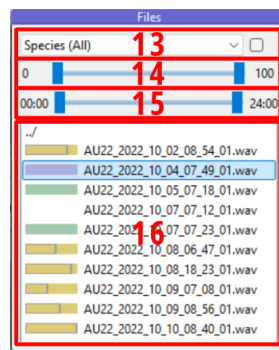


Figure 5: The Files panel of the main interface.

- (13) A dropdown menu containing all the species for which annotations are found in the current directory. Choosing a species from the list will also adjust the file icons in

the list. Ticking the box next to the dropdown menu will shrink the file list to files containing annotations with the selected species.

- (14) These confidence sliders can be used to filter the file list for confidence values.
- (15) These time sliders can be used to only select recordings from the specified time frame. That only works for files and if the filename contains date and time in the format `<year><month><day>_<hour><minute><second>`.
- (16) The list of files. Next to the files names a file icon indicates the status of the files. It contains information on the status of the annotations in the file. See 5.5 for details.

9.1.4 Controls

The Controls panel is connected to the spectrogram and offers different options to play sound, adjust the spectrogram and handle the currently selected segment.

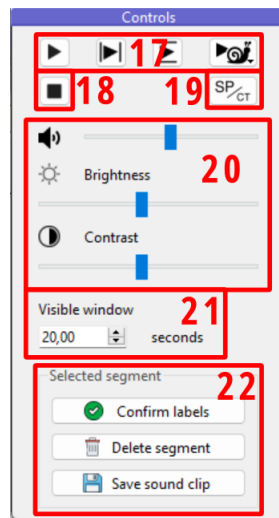


Figure 6: The Controls panel of the main interface.

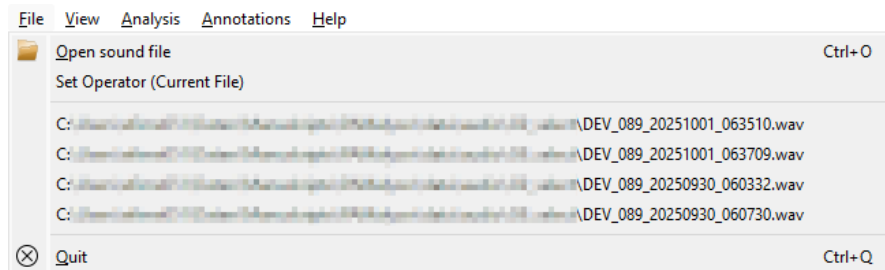
- (17) Different buttons to play back sound. From left to right they are:
 - Start/continue playback of the visible section (shortcut **Space**). The button turns into a *Pause* button while playing.
 - Play all frequencies of the selected segment (▶, shortcut: **Ctrl + Space**).
 - Play highlighted frequencies of the selected segment (▶|).
 - Play selected segment at different speeds (▶⚙). Hold the button down to change it to 2x (twice as fast) or 1/2 or 1/4 speed. Note that the pitch also changes.

- (18) Stop playback and have the slider return to the start of the visible section.
- (19) Change between species and call type labels (shortcut **Tab**).
- (20) 3 sliders:
 - Change the volume of playback using the slider below the buttons.
 - Change brightness of the spectrogram.
 - Change contrast of the spectrogram.
- (21) Change the length of the visible window of the audio file (highlighted blue in the overview spectrogram (3)).
- (22) 3 buttons to handle the selected segment:
 - Set confidence to 100% for all the species labels of the selected segment (shortcut **Return**).
 - Delete the selected segment (shortcut **Backspace/Del**).
 - Save the selected segment as sound file.

9.2 Menu

Most of the options for PAMalyzer are found through the menus. There are keyboard shortcuts for many of the menu items, which can be seen in the menus themselves. You can find a reference of shortcuts in the section [Keyboard shortcuts](#)

9.2.1 File Menu



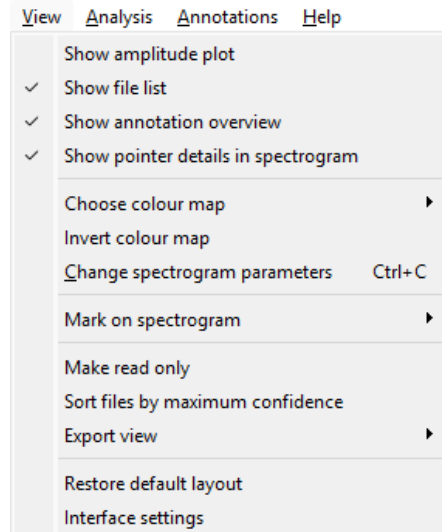
Open sound file Produces a file dialog so that you can choose a new audio file to open.

Set operator (Current File) Enables you to change the operator or reviewer that were specified when you started PAMalyzer. The change only applies to this file.

List of recently used files Makes it easy to revisit recent files.

Quit Does what it says.

9.2.2 View Menu



Changing appearance The first four options in the menu hide or reveal the amplitude plot, file list, annotation overview, and information about where the mouse is pointing in the spectrogram. The ones that are selected are marked with a tick.

Choose colour map This enables the user to select a color map for the spectrograms they prefer to the default grey one. To adjust the settings for the custom color map, use the option *Interface settings* on the bottom of the menu.

Invert colour map By default, areas of high energy (frequencies where there is a call or other sound) are shown as the lightest color, and low energy as dark. This can be swapped over with this option. Note that you will need to change the brightness and contrast using the [Controls](#) area after inverting the color map.

Change spectrogram parameters A set of options to modify the spectrogram. See the section [Changing the Spectrogram Computation](#) for more details.

Mark on spectrogram There are some features that will help you spot bird calls or recognise them. We currently offer two options here: the fundamental frequency and points of high energy. To select one, choose it from the menu; select it again to remove it. They can help to find particular types of call.

Make read only When reviewing a segmentation it is possible to click on the plots by mistake, adding further segments. This option avoids this problem. It can be turned off by selecting it again. When read only mode is on, the message section at the bottom of the screen says so.

Sort files by maximum confidence This option sorts the list of files by the maximum confidence value of detections for the species selected in the *Files* section. To enable a rank-based review process it excludes confirmed segments from sorting.

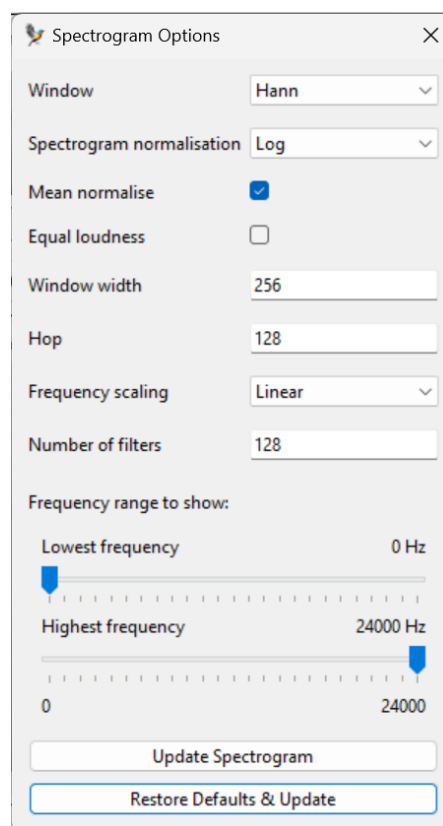
Export view Either export the overview spectrogram (from the overview panel) or the detailed spectrogram (from the spectrogram panel, including annotations) as image.

Restore default layout Returns the various screen components to their original layout if they have been moved around.

Interface settings This option produces a dialog that enables the user to customize several things about PAMalyzer. See the section [Interface settings](#) for more details.

Changing the Spectrogram Computation

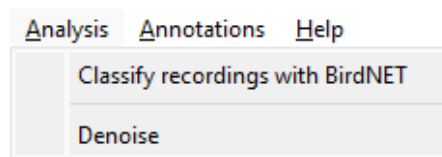
- Selecting *Change spectrogram parameters* from the *View* menu (or the shortcut **Ctrl + C**) will produce the following dialog box:



- You can change some of the parameters that are used to produce the spectrogram:
 - The window function (default: Hann) controls how the spectrogram combines sounds across the time range.
 - Mean normalization and equal loudness try to make the spectrogram energies more even.

- The window width and hop size control how much of the audio file makes up one spectrogram bin, and how they overlap. A big window size will improve frequency resolution, but reduce time resolution, and vice versa. The hop size controls how much the spectrogram time bins overlap. It is computationally more efficient if these numbers are powers of 2 (such as 256, 512, 1024).
- The frequency range sliders let you change the visible frequency range in the main spectrogram window. This can be useful if you want to focus on only part of the range.

9.2.3 Analysis Menu



These options will mostly provide dialog boxes that ask you to make choices.

Classify recordings with BirdNET Choosing this option a window will open where you can set the parameters for BirdNET to classify all your recordings in the current directory and in all subdirectories. It is described in more detail in the section [Analyze Recordings with BirdNET](#)

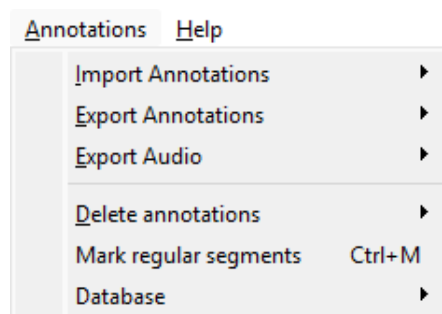
Denoise Runs some programs that try to get rid of the noise in the audio file so that the bird calls are easier to see. You can save the denoised audio files, and undo the denoising if it does not help.

Wavelets This is our main method, and tries to preserve the bird call perfectly.

Bandpass Suppresses all frequencies outside a restricted range (which you specify) so that you can concentrate on the frequency range where calls that you are interested in can be seen.

Butterworth bandpass Another way to compute a restricted frequency range.

9.2.4 Annotations Menu



Import Annotations Offers options to import annotations from other formats:

From AviaNZ (.data files) If you have made user annotations in other programs that you want to import, this option will help to do it.

From Raven (selection tables) If you have made user annotations in other programs that you want to import, this option will help to do it.

Export Annotations Options to export annotations to other formats:

To Excel (File) This enables the user to save a summary of the annotation of the currently opened file into an Excel workbook (in the same location as the current audio file), as is described in section [Export](#).

To Excel (Directory) This option summarizes all the annotations of the current directory (including subdirectories) into Excel workbooks as described in section [Export](#).

To database If you want to copy a set of user annotations (for example to have them backed up, or to transfer them to another computer) without the wav files, but with the directory structure preserved, this option will do that.

To .data files Export to the old AviaNZ format.

Export Audio Options to export audio segments:

Files with selected species Copies all files containing the selected species to the given directory. Only work if a species is selected. The confidence filter is applied as well.

File segments with selected species Extracts all audio segments with the selected species. Creates a subdirectory in the given directory for the species. Also works when no species is selected. The confidence filter is applied as well.

Delete Annotations Options to delete annotations:

File annotations Deletes all annotations from the currently open file

Directory annotations Deletes all annotations in the currently open directory

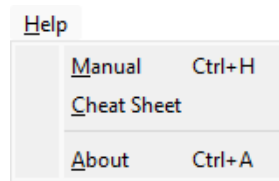
Mark regular segments Some users find it helpful to have an automatically generated segment box at regular intervals, for example to label all the birds in that segment under some statistical protocol. By default these are boxes are made to be 15 seconds long, with one box every 300 seconds (5 minutes) through the file. These options can be changed using the *Check-Ignore protocol* area in the *Interface settings*. There is also an option there to simply mark those areas rather than putting a segment box in place.

Database Options for database handling.

Choose database Choose a database for storing annotations. Either choose an existing one or type a name to create a new one.

Update file directory Scans the currently open directory and sets the directory in the database for all the files within it. Use this if you moved files or directories.

9.2.5 Help Menu



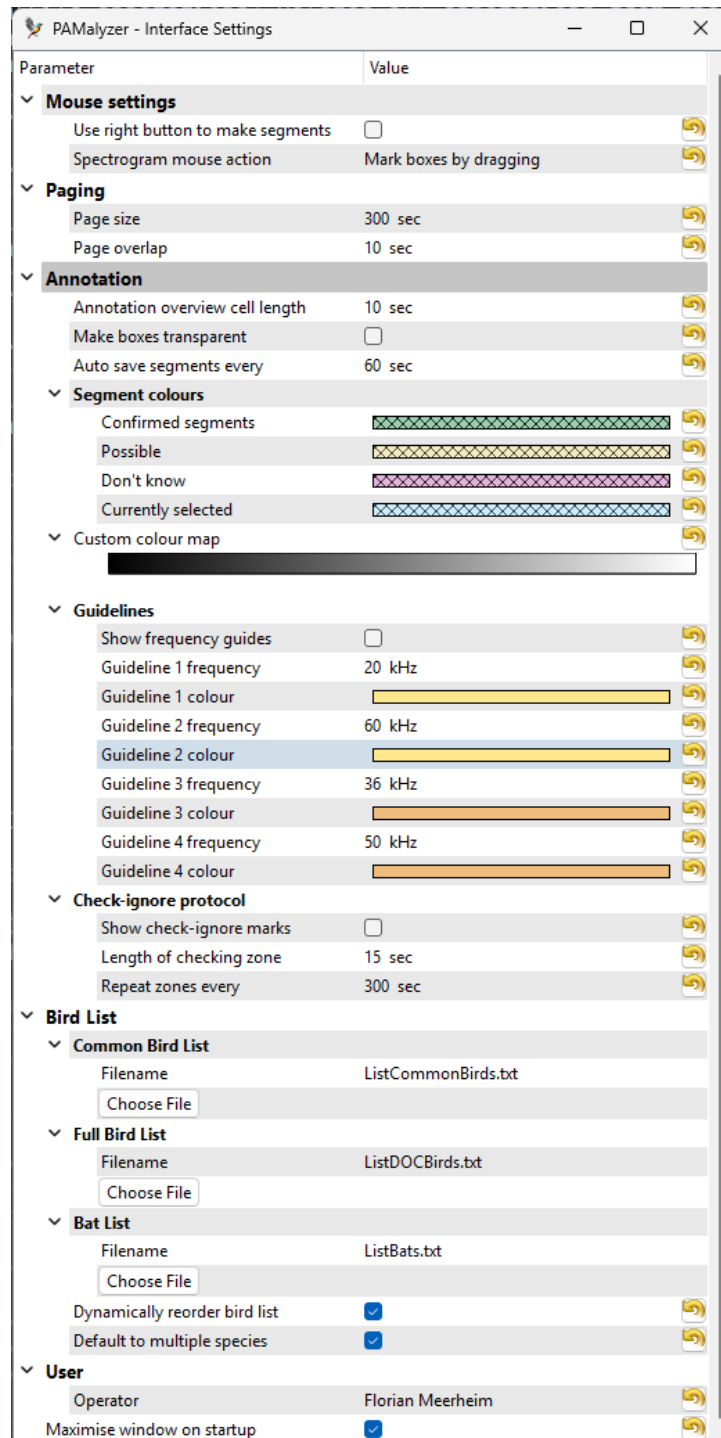
Help Gives access to this manual.

Cheat sheet Gives access to a cheat sheet with useful shortcuts and some information about the BirdNET classifier parameters.

About Shows you basic information about PAMalyzer.

9.3 Interface settings

- There are quite a few user-selectable options in PAMalyzer, which you can choose using the *Interface settings* menu option in the *View* menu, which will produce the following dialog box:



- These options include:

Mouse settings Swap which mouse button selects segments and which creates them, and change the method of creating segments (described in detail in

[Create and Label Segments](#)).

Paging The page is the length of spectrogram loaded into PAMalyzer at one time. By default it is 300 seconds (5 minutes). Note that longer times will require more memory and processing time. The amount of overlap between the pages can also be specified. The aim of the overlap is to make sure that calls aren't missed, and that segments are labeled accurately, at the page limits.

Annotation – The first option in this section changes the length of the boxes in the Overview.

- The next enables you to make the labeling boxes transparent (so that only the outline of the box is shown) if you find it hard to see what is in a segment.
- By default, PAMalyzer save the segments you have made every 60 seconds. This can be changed, for example if you want to do it more often for safety.
- You can also change the colors of segments and apply a custom color map.
- PAMalyzer has a check-ignore protocol for people who only annotate subsets of recordings. This puts a mark on the spectrogram in places where the user should be annotating the spectrogram, which you can control with these options.

Bird list There are two bird lists in PAMalyzer: the short list of common birds, and a longer list of all possibilities, together with a bat list. You can change these files for others (for example, to include non-New Zealand species).

You can also change the default behaviors of dynamically updating the list of common birds, and enable multiple species to be selected for a single segment. This can be useful for labeling things like the dawn chorus, but these segments will not provide good training data for new recognisers.

User Enables you to set the operator and reviewer; this facility can also be found in the *File* menu. You can also change whether or not the PAMalyzer window starts at full screen size.

9.4 Keyboard shortcuts

Action	Shortcut
General	
Open audio file	Ctrl + O
Quit program	Ctrl + Q
Open the Manual	Ctrl + H
Show About Info	Ctrl + A
Navigation	
Open the previous audio file in the file list	↑

Open the next audio file in the file list	↓
Move backward within the current audio file	←
Move forward within the current audio file	→
Jump to the next page (page size set in Interface settings)	Shift + →
Jump to the previous page	Shift + ←
Jump to the next segment	Ctrl + →
Jump to the previous segment	Ctrl + ←
Navigate to the previous file including hidden files	Alt + ↑
Navigate to the next file including hidden files	Alt + ↓
Playback	
Play / pause the visible spectrogram area	Space
Play only the selected segment	Ctrl + Space
Segment creation	
Apply the label of the previously created segment to the new one (no menu shown)	Shift + drag
Mark the new segment as uncertain (adds a ? to the label, confidence of 50%)	Ctrl + drag
Editing	
Confirm the label of the selected segment	Enter
Toggle between Species and Call-type view (SP/CT button)	Tab
Delete the currently selected segment	Delete/Backspace
Delete all segments in the current file	Ctrl + D
Spectrogram	
Change spectrogram parameters	Ctrl + C
Export current view as image	Ctrl + I